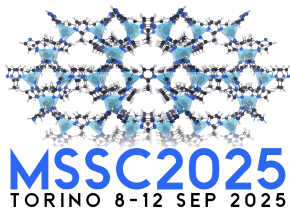


THE PERIODIC LCAO APPROACH

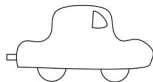
S. Casassa

Università degli Studi di Torino (Italy) & Imperial College London (UK)



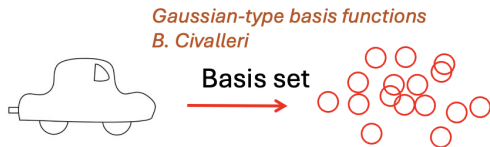
COMPUTE THE PROPERTIES OF A SYSTEM

WHAT DOES IT MEAN?



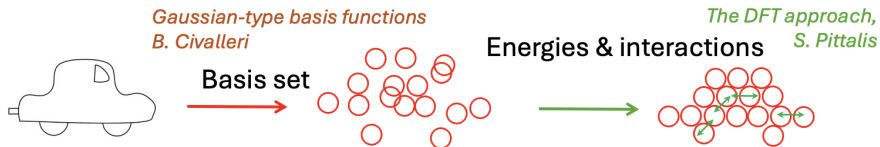
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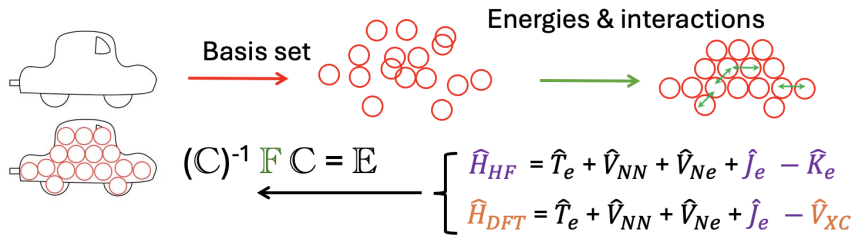


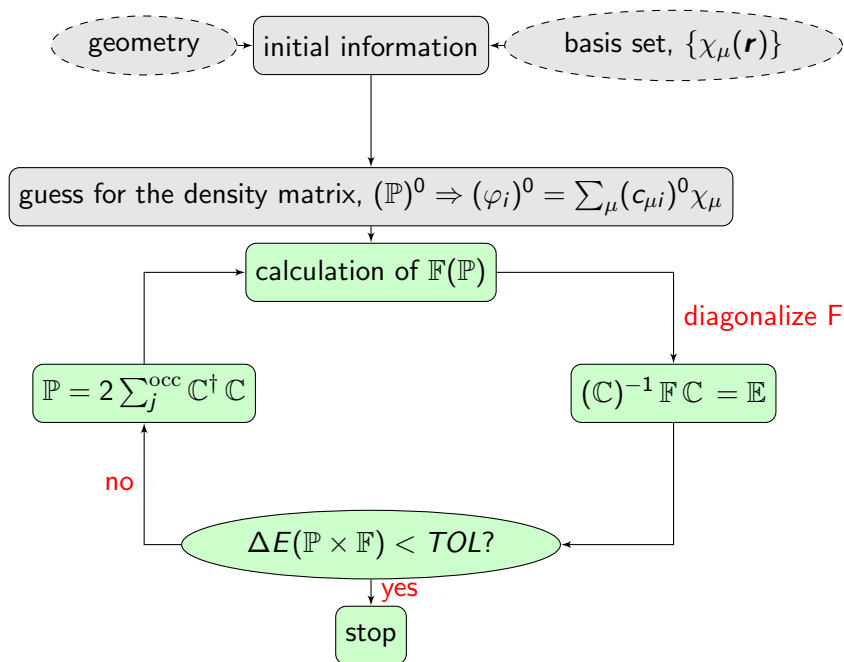
$$\hat{H}_{HF} = \hat{T}_e + \hat{V}_{NN} + \hat{V}_{Ne} + \hat{J}_e - \hat{K}_e$$

$$\hat{H}_{DFT} = \hat{T}_e + \hat{V}_{NN} + \hat{V}_{Ne} + \hat{J}_e - \hat{V}_{XC}$$

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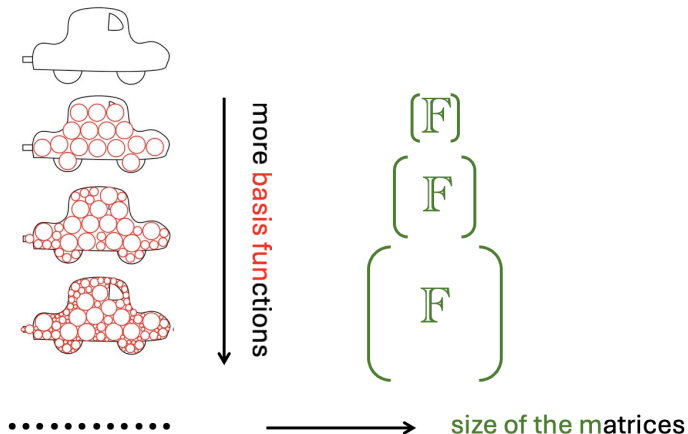
WHAT DOES IT MEAN?



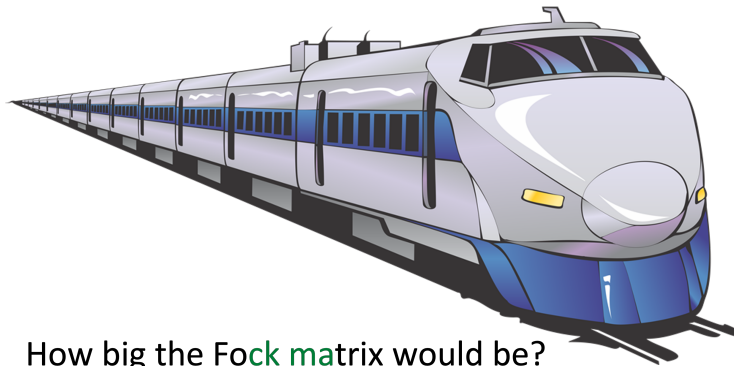


COMPUTE THE PROPERTIES OF A SYSTEM

WHAT DOES IT MEAN?

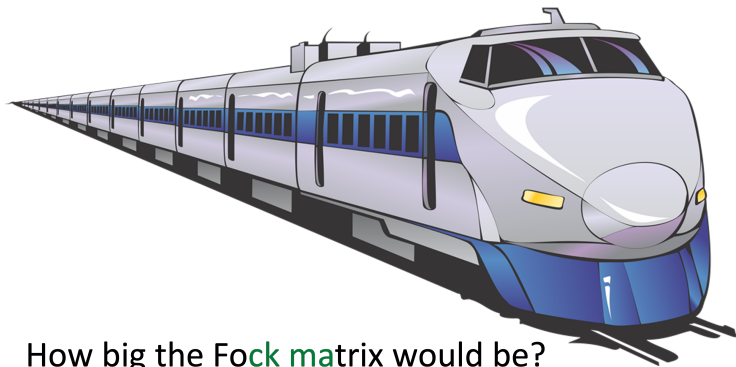


COMPUTE A MACROSCOPIC SYSTEM?



How big the Fock matrix would be?

COMPUTE A MACROSCOPIC SYSTEM?



How big the Fock matrix would be?

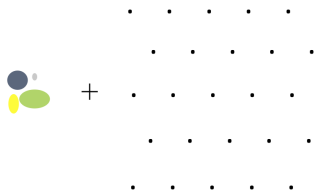
In condensed matter, **symmetry matters**,
...and **must be exploited** at different levels and
in different spaces!

CRYSTALLOGRAPHY



Basis

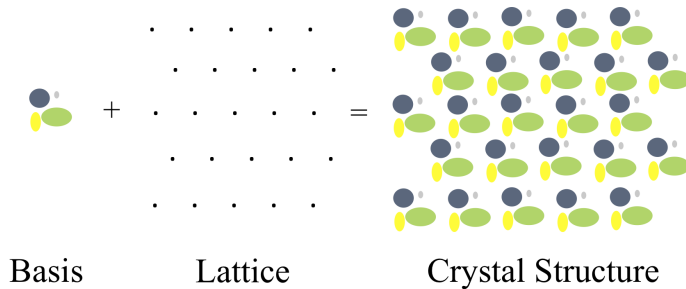
CRYSTALLOGRAPHY



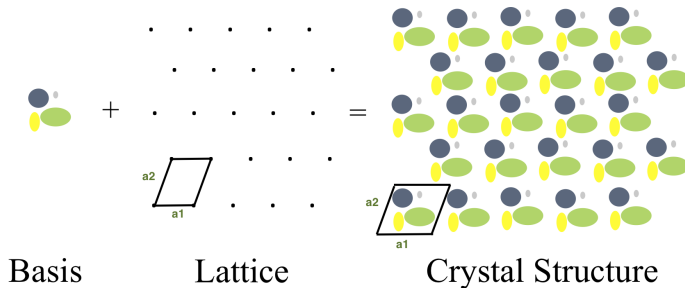
Basis

Lattice

CRYSTALLOGRAPHY

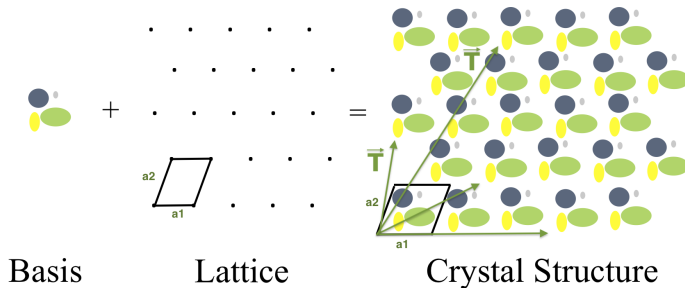


CRYSTALLOGRAPHY



Lattice parameters: $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ and α, β, γ

CRYSTALLOGRAPHY



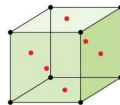
Lattice parameters: $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ and α, β, γ

Lattice vectors: $\{\hat{\mathbf{T}}\}$

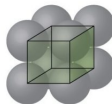
Systems	lattice parameters	
Triclinic	$a_1 \neq a_2 \neq a_3$ $\alpha_{12} \neq \alpha_{23} \neq \alpha_{31}$	
Monoclinic	$a_1 \neq a_2 \neq a_3$ $\alpha_{23} = \alpha_{31} = 90^\circ$ $\alpha_{12} \neq 90^\circ$	
Orthorhombic	$a_1 \neq a_2 \neq a_3$ $\alpha_{12} = \alpha_{23} = \alpha_{31} = 90^\circ$	
Tetragonal	$a_1 = a_2 \neq a_3$ $\alpha_{12} = \alpha_{23} = \alpha_{31} = 90^\circ$	
Trigonal	$a_1 = a_2 = a_3$ $\alpha_{12} = \alpha_{23} = \alpha_{31} < 120^\circ$	
Cubic	$a_1 = a_2 = a_3$ $\alpha_{12} = \alpha_{23} = \alpha_{31} = 90^\circ$	
Hexagonal	$a_1 = a_2 \neq a_3$ $\alpha_{12} = 120^\circ$ $\alpha_{23} = \alpha_{31} = 90^\circ$	

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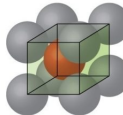
Lattice point locations



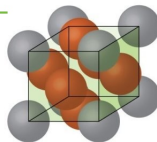
Cubic unit cells



Simple cubic



Body-centered cubic



Face-centered cubic

Bravais lattice	Parameters	Simple (P)	Volume centered (I)	Base centered (C)	Face centered (F)
Triclinic	$a_1 \neq a_2 \neq a_3$ $\alpha_{12} \neq \alpha_{23} \neq \alpha_{31}$				
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14 Bravais Lattice



230 Space Groups



32 Point Symmetry Groups

REAL CRYSTALS

- ☹ has an ∞ number of atoms and electrons
- ☹ contains **defects**, i.e. dislocations, vacancies, interstitial and ertro atoms, etc.. .
- ☹ has **surfaces**

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↪ Translational invariance and long-range order are almost **never satisfied**



SYMMETRIES

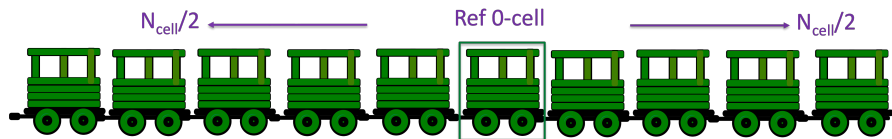
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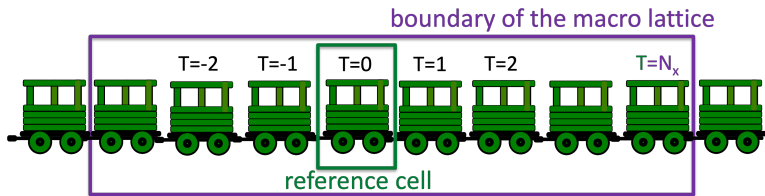
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SYMMETRIES

- ☺ if **translational invariance** is imposed
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- 🌿 a **macro-lattice** of identical $N_{\text{cell}} = N_x \times N_y \times N_z$ cells can be defined
- 🌿 and the **size** of the macro-lattice is defined by **thresholds on integrals**:
 $\langle \chi_\mu(\mathbf{r}) \chi_\nu(\mathbf{r}) | | \chi_\lambda(\mathbf{r}) \chi_\rho(\mathbf{r}) \rangle$

1. POINT SYMMETRY

DIRECT SPACE

 given the **space group** $\{\hat{R}\}$ and the info on the **asymmetric unit**:

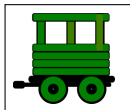
- build up the infinite system (see the case of nanotubes!)
- calculation of the **irreducible integrals**
- calculation of the **irreducible part of the matrices**

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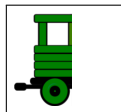
🚂 given the **space group** $\{\hat{R}\}$ and the info on the **asymmetric unit**:

reducible number of objects



$\leftarrow \{\hat{R}\}$

irreducible



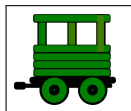
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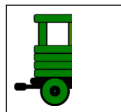
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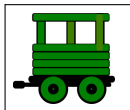
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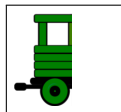
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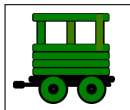
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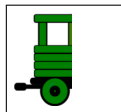
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EXAMPLES

MATRICES SIZE	P(G)	2600784	F(G)	1302232
	P(G) IRR	139064	F(G) IRR	139064

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SLAB

77

2.47

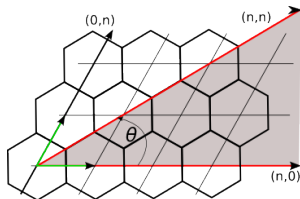
1

6 0.3333 0.6667 0.0000

NANOTUBE

9 3

END



EXAMPLES

MATRICES SIZE

P(G)	2600784	F(G)	1302232
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SLAB

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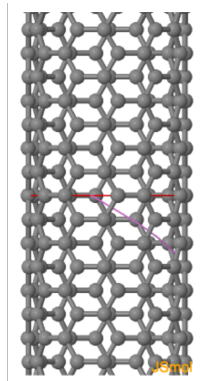
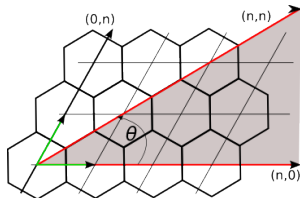
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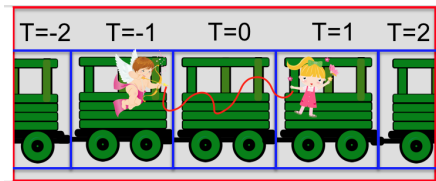
END



2. TRANSLATIONAL SYMMETRY

IN DIRECT SPACE

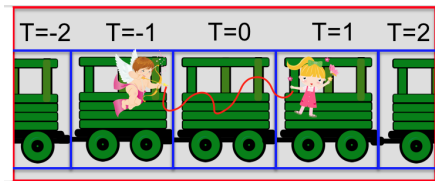
 given the **translational invariance** of the system



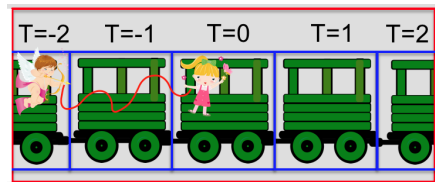
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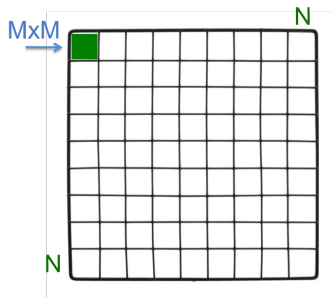
↔ all the contributions to the Fock matrix are calculated with respect to the **reference cell**



A CLOSER LOOK AT THE FOCK MATRIX

IN DIRECT SPACE

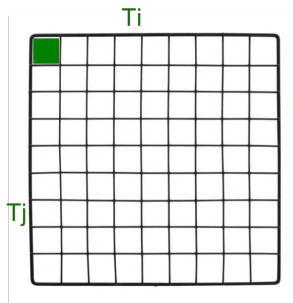
- Representation in the **AO basis set**
- a set of **M** AO: $\{\chi_{\mu}^0(\mathbf{r})\}^M$ in the reference cell
- dimension of the system: **N** cells
- Square matrix: **$N \times N$** blocks of size **M**



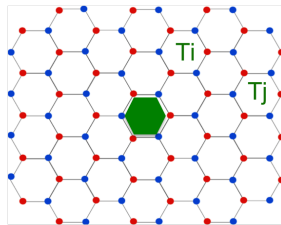
A CLOSER LOOK AT THE FOCK MATRIX

IN DIRECT SPACE

Fock matrix in the AO basis set



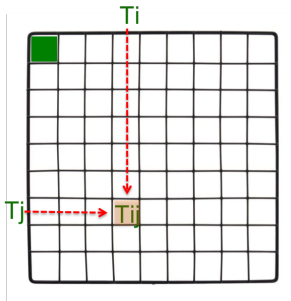
Direct hexagonal 2D lattice



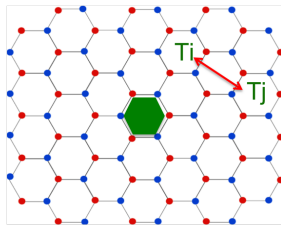
A CLOSER LOOK AT THE FOCK MATRIX

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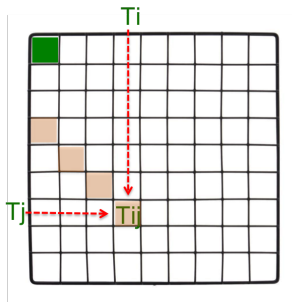


$$\langle \chi_{\mu}^{\mathbf{T}_i}(\mathbf{r}) | \hat{F} | \chi_{\nu}^{\mathbf{T}_j}(\mathbf{r}) \rangle =$$

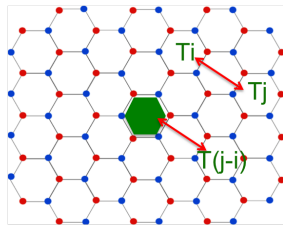
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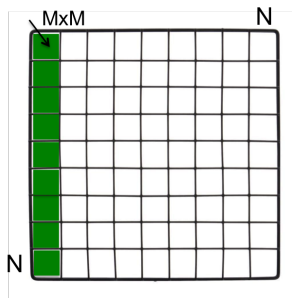


$$\langle \chi_{\mu}^{\mathbf{T}_i}(\mathbf{r}) | \hat{F} | \chi_{\nu}^{\mathbf{T}_j}(\mathbf{r}) \rangle = \langle \chi_{\mu}^{\mathbf{0}}(\mathbf{r}) | \hat{F} | \chi_{\nu}^{(\mathbf{T}_j - \mathbf{T}_i)}(\mathbf{r}) \rangle \equiv F_{\mu\nu}^{\mathbf{T}_{ji}}$$

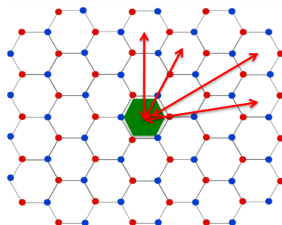
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3. TRANSLATIONAL SYMMETRY

BLOCH'S THEOREM

In a crystal with a **periodic potential**, $V(\mathbf{r}) \equiv V(\mathbf{r} + \mathbf{T})$, the eigenvectors solution of the Schrödinger equation, i.e. the crystalline orbitals, $\Psi_i(\mathbf{r})$, must take the form of **Bloch functions**

$$\Psi_i^{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\mathbf{r}} u_i(\mathbf{r})$$

where $u_i(\mathbf{r})$ has the same **periodicity of the direct lattice** $u_i(\mathbf{r}) = u_i(\mathbf{r} + \mathbf{T})$ and $e^{i\mathbf{k}\mathbf{r}}$ is a plane wave.

Crystalline Orbitals (CO) are Bloch Functions (BF)

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How to get Bloch functions?



Nothing could be easier ..if only we had chosen a basis set of plane waves



Coffee-break



Coffee-break

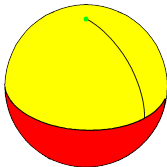
from “real” to “model”





Coffee-break

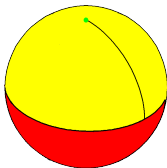
from “real” to “model”





Coffee-break

from “real” to “model”



from molecule to lattice

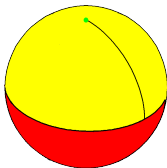
$$\chi_{\mu}(r)$$





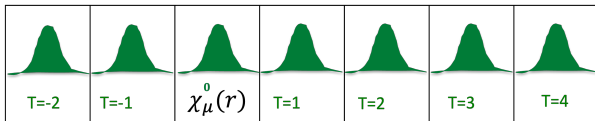
Coffee-break

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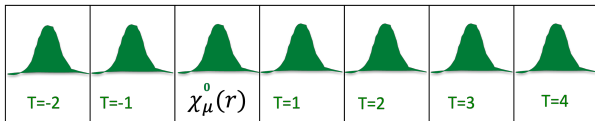
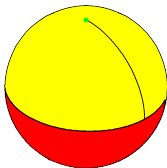
Coffee-break

from “real” to “model”



from molecule to lattice

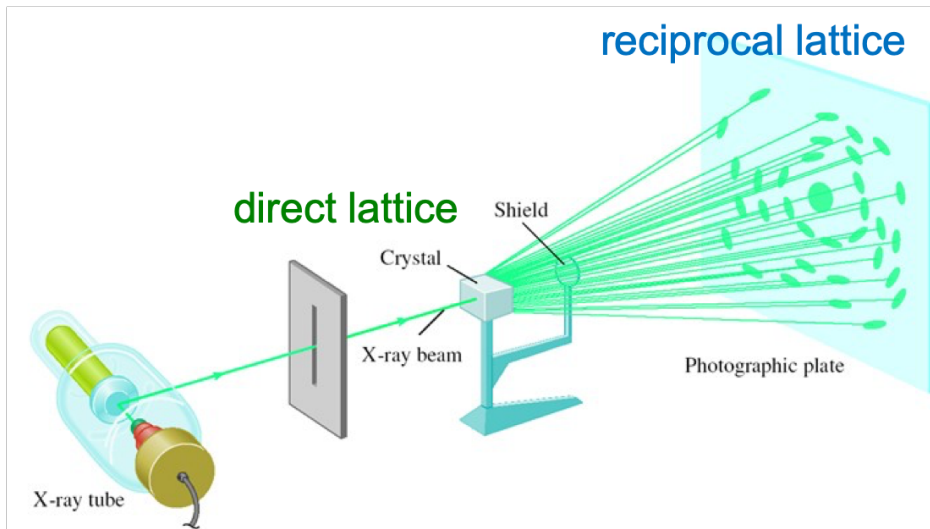
$$\chi_{\mu}(r)$$



And a big question:

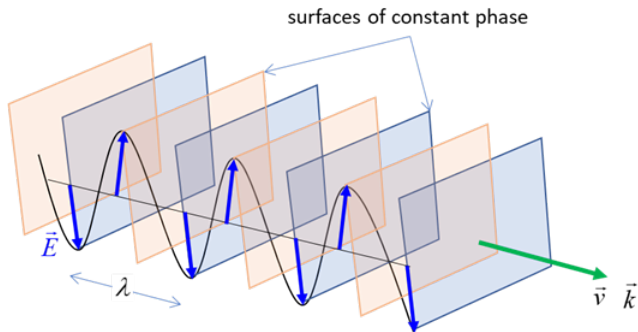
how to get crystalline orbitals out of atomic orbitals?

WORKING BETWEEN TWO LATTICES



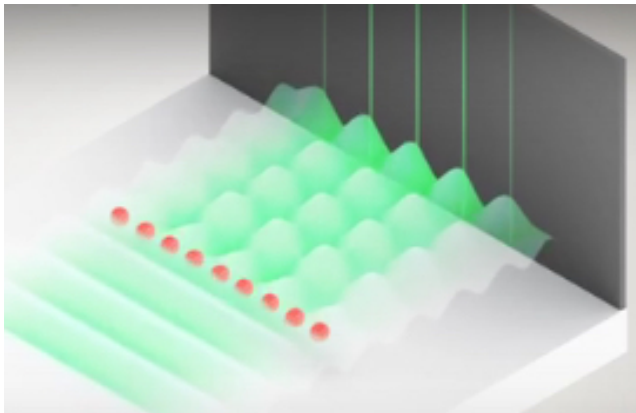
WHAT'S RECIPROCAL LATTICE?

THE SET OF ALL WAVE VECTORS $\mathbf{k}$ THAT
yield plane waves with the same periodicity of a given Bravais lattice,
referred as **direct lattice**, forms a lattice known as **reciprocal lattice**.



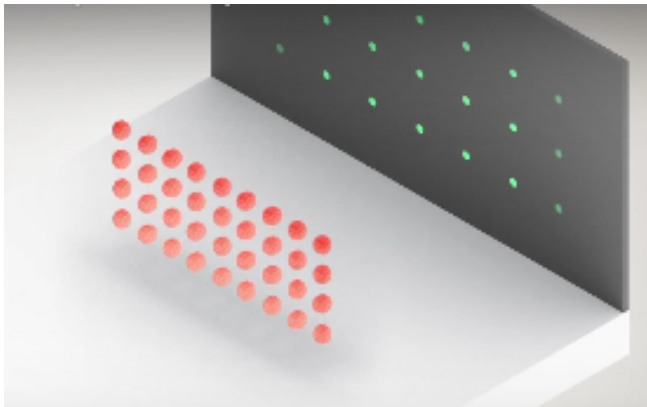
WHAT'S RECIPROCAL LATTICE?

- 🔍 when X-rays interact with a periodic array of atoms, the pattern observed is a periodic array of lines



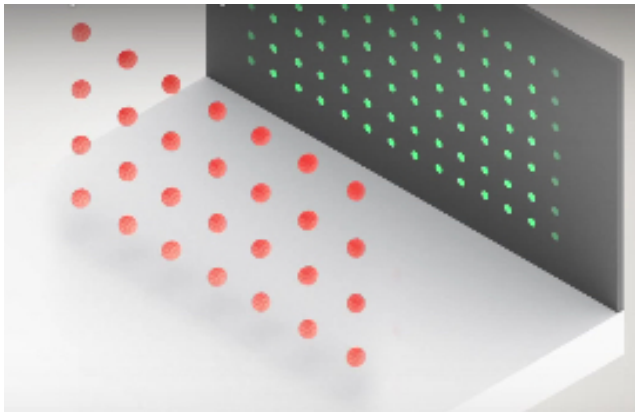
WHAT'S RECIPROCAL LATTICE?

 for a crystal, the diffraction pattern is a regular array of light spots, arranged in a **reciprocal lattice**



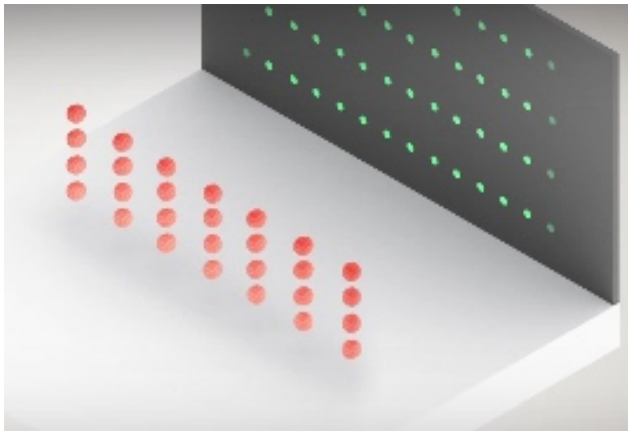
WHAT'S RECIPROCAL LATTICE?

 if atoms are more widely spaced, light spots get closer



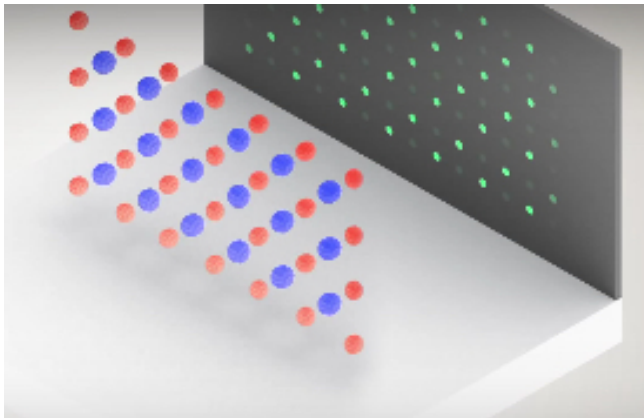
WHAT'S RECIPROCAL LATTICE?

 when atoms get closer, the reciprocal lattice expands



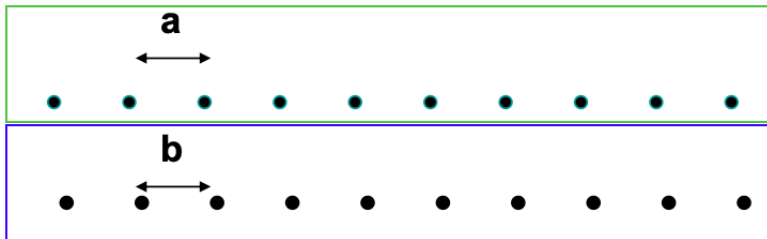
WHAT'S RECIPROCAL LATTICE?

 if other atoms are added, the intensity of the light spots may vary



DIRECT $\rightarrow$ RECIPROCAL

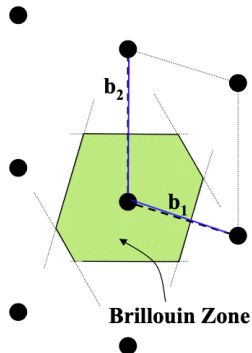
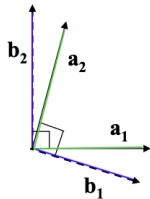
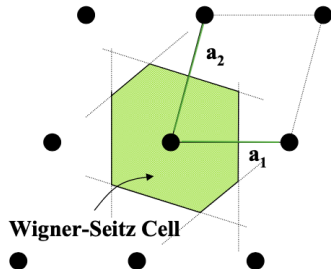
LATTICE IN 1D



$$b = 2\pi a$$

DIRECT $\rightarrow$ RECIPROCAL

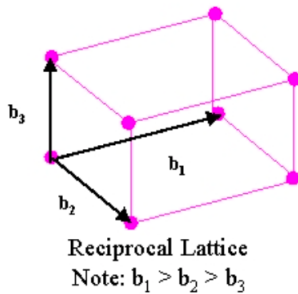
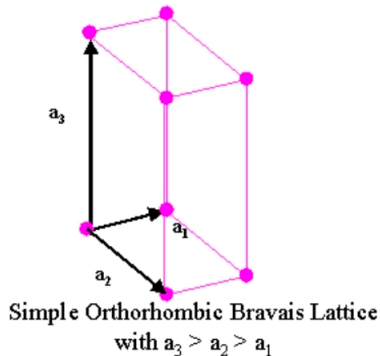
LATTICE IN 2D



$$\mathbf{b}_i = \frac{2\pi}{\mathbf{a}_j} \delta_{ij}$$

DIRECT $\rightarrow$ RECIPROCAL

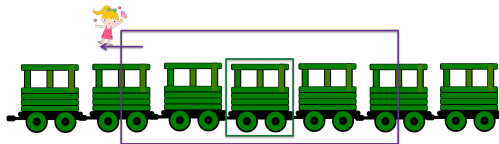
LATTICE IN 3D



$$\mathbf{b}_i = \frac{2\pi}{|\mathbf{a}_i(\mathbf{a}_j \times \mathbf{a}_k)|} (\mathbf{a}_j \times \mathbf{a}_k)$$

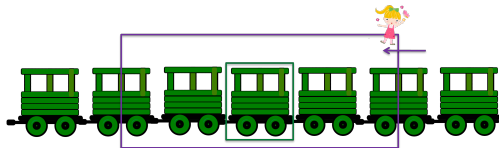
CYCLIC BOUNDARY CONDITIONS

∞ Everything that comes out from the left comes back in from the right:
the crystal is limited but **infinite**



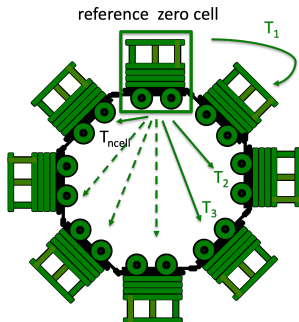
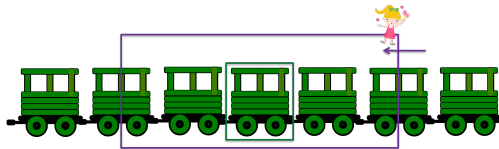
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CYCLIC BOUNDARY CONDITIONS

∞ Everything that comes out from the left comes back in from the right:
the crystal is limited but **infinite**



i.e. in the case of a periodic potential

$$V(\mathbf{r}) = V(\mathbf{r} + \mathbf{T})$$

the **translation operators**, $\{\hat{\mathbf{T}}\}$ form the group of the Schroedinger equation, $[\hat{H}, \hat{\mathbf{T}}]$, which is an **abelian CYCLIC** group

EXPLOIT CYCLIC BOUNDARIES

(1) Given N_{cell} , the character Table is computed by finding the $N_{\text{symm op}} \times N_{\text{IRREP}}$ roots of the unity

N_{irrep}	T_0		T_i		T_j
k_0	1		1		1
k_1	1		.		.
.	.		.		.
.	.		.		.
k_m	1		$\exp(i K_m T_i)$		$\exp(i K_m T_j)$
.	.		.		.
.	.		.		.
$K_{\text{ncell-1}}$	1		$\exp(i K_n T_i)$		$\exp(i K_n T_j)$

EXPLOIT CYCLIC BOUNDARIES

(1) Given N_{cell} , the character Table is computed by finding the $N_{\text{symm op}} \times N_{\text{IRREP}}$ roots of the unity

N_{irrep}	T_0		T_i		T_j
k_0	1		1		1
k_1	1		$\vdots$		$\vdots$
$\vdots$	$\vdots$		$\vdots$		$\vdots$
k_m	1		$\exp(i K_m T_i)$		$\exp(i K_m T_j)$
$\vdots$	$\vdots$		$\vdots$		$\vdots$
$K_{\text{n cell}-1}$	1		$\exp(i K_n T_i)$		$\exp(i K_n T_j)$

(2) For each IRREP, Projector operators can be defined

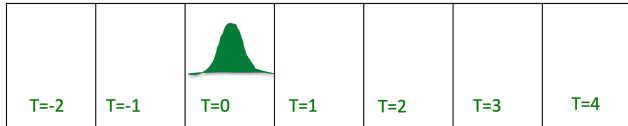
$$\hat{P}^k = \frac{1}{\sqrt{N_{\text{cell}}}} \sum_T^{N_{\text{cell}}} e^{i \mathbf{k} \cdot \mathbf{T}} \hat{T}$$

EXPLOIT CYCLIC BOUNDARIES..TO BE CONTINUED

(3) Bloch functions can be obtained by applying the Projector operators, $\hat{P}^{\mathbf{k}}$, to the atomic orbitals of the reference cell, $\{\chi_{\mu}^0\}^M$

$$\hat{P}^{\mathbf{k}} \chi_{\mu}^0(\mathbf{r}) = \frac{1}{\sqrt{N_{\text{cell}}}} \sum_{\mathbf{T}} e^{i\mathbf{k} \cdot \mathbf{T}} \hat{T} \chi_{\mu}(\mathbf{r})$$

AO
 $\{\chi_{\mu}^0\}^M$

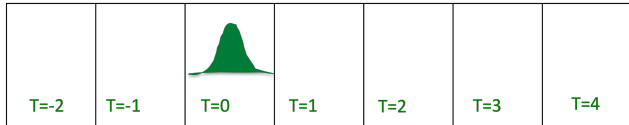


EXPLOIT CYCLIC BOUNDARIES..TO BE CONTINUED

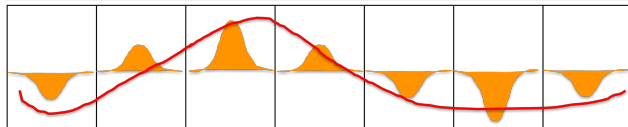
(3) Bloch functions can be obtained by applying the Projector operators, $\hat{P}^{\mathbf{k}}$, to the atomic orbitals of the reference cell, $\{\chi_{\mu}^0\}^M$

$$\hat{P}^{\mathbf{k}} \chi_{\mu}^0(\mathbf{r}) = \frac{1}{\sqrt{N_{\text{cell}}}} \sum_{\mathbf{T}} e^{i\mathbf{k} \cdot \mathbf{T}} \hat{T} \chi_{\mu}(\mathbf{r}) \equiv \frac{1}{\sqrt{N_{\text{cell}}}} \sum_{\mathbf{T}} e^{i\mathbf{k} \cdot \mathbf{T}} \chi_{\mu}^{\mathbf{T}}(\mathbf{r}) \equiv \Phi_{\mu}^{\mathbf{k}}(\mathbf{r})$$

AO
 $\{\chi_{\mu}^0\}^M$



BF
 $\{\Phi_{\mu}^{\mathbf{k}}(\mathbf{r})\}$



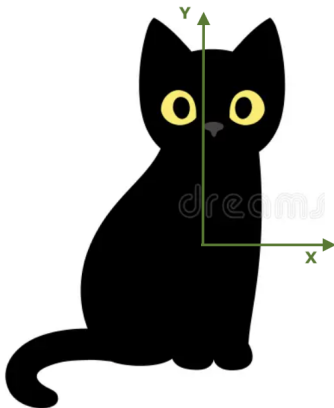


Coffee-break



Coffee-break

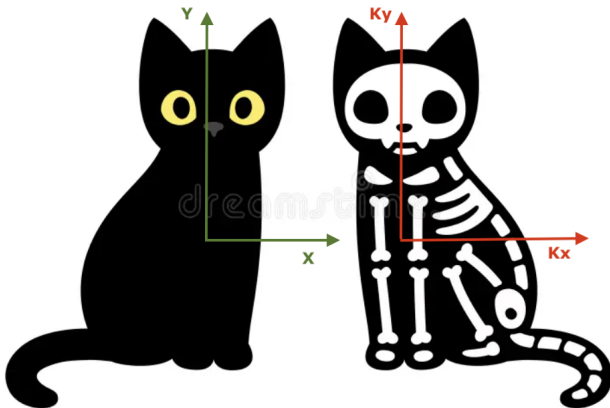
coordinate space, $[\{\mathbf{R}\}, \{\mathbf{r}\}]$





Coffee-break

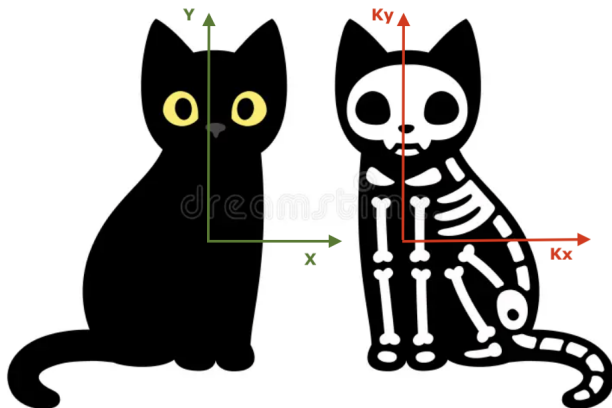
coordinate space, $[\{\mathbf{R}\}, \{\mathbf{r}\}]$ plus momentum space, $[\{\mathbf{K}\}, \{\mathbf{k}\}]$





Coffee-break

coordinate space, $[\{\mathbf{R}\}, \{\mathbf{r}\}]$ plus momentum space, $[\{\mathbf{K}\}, \{\mathbf{k}\}]$



And a big question:

What about the Fock matrix in the basis set of Bloch functions?

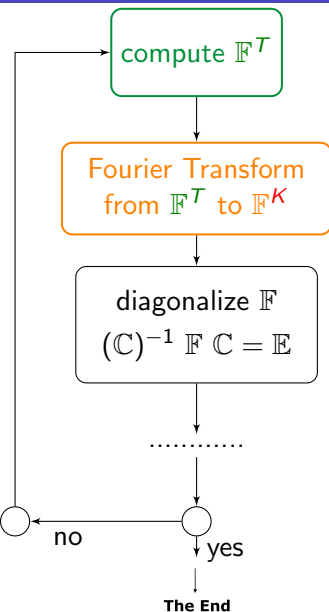
Exploit orthogonality

- Fock matrix in the BFs basis set, i.e. from direct to reciprocal space

$$F_{\mu\nu}^{k_1, k_2} = \langle \phi_{\mu}^{k_1}(\mathbf{r}) | \hat{F} | \phi_{\nu}^{k_2}(\mathbf{r}) \rangle = \sum_{\mathbf{T}} e^{i\mathbf{k} \cdot \mathbf{T}} F_{\mu\nu}^T$$

- Fock matrix elements between functions belonging to different IRREP are null

$$F_{\mu, \nu}^{k_1, k_2} = \langle \phi_{\mu}^{k_1}(\mathbf{r}) | \hat{F} | \phi_{\nu}^{k_2}(\mathbf{r}) \rangle = F_{\mu, \nu}^{k_1} \delta_{k_1, k_2}$$



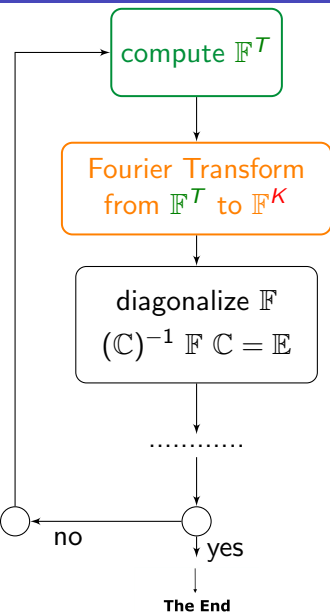
Exploit orthogonality

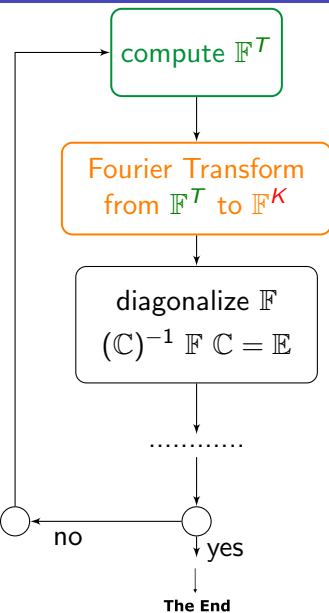
- Fock matrix in the BFs basis set, i.e. from direct to reciprocal space

$$F_{\mu\nu}^{\mathbf{k}_1, \mathbf{k}_2} = \langle \phi_{\mu}^{\mathbf{k}_1}(\mathbf{r}) | \hat{F} | \phi_{\nu}^{\mathbf{k}_2}(\mathbf{r}) \rangle = \sum_{\mathbf{T}} e^{i\mathbf{k} \cdot \mathbf{T}} F_{\mu\nu}^{\mathbf{T}}$$

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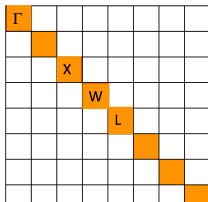
- Fock matrix in the BFs basis set,
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$$F_{\mu\nu}^{\mathbf{k}_1, \mathbf{k}_2} = \langle \phi_{\mu}^{\mathbf{k}_1}(\mathbf{r}) | \hat{F} | \phi_{\nu}^{\mathbf{k}_2}(\mathbf{r}) \rangle = \sum_{\mathbf{T}} e^{i\mathbf{k} \cdot \mathbf{T}} F_{\mu\nu}^{\mathbf{T}}$$

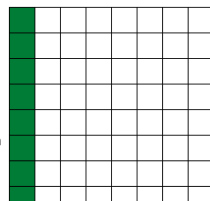
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$$F_{\mu,\nu}^{\mathbf{k}_1,\mathbf{k}_2} = \langle \Phi_{\mu}^{\mathbf{k}_1}(\mathbf{r}) | \hat{F} | \Phi_{\nu}^{\mathbf{k}_2}(\mathbf{r}) \rangle = F_{\mu,\nu}^{\mathbf{k}_1} \delta_{\mathbf{k}_1,\mathbf{k}_2}$$

Fock Matrix in **BFs** basis
(MxM) x **N**_{IRREP}



Fock Matrix in **AOs** basis
(MxM) x T_{cell}



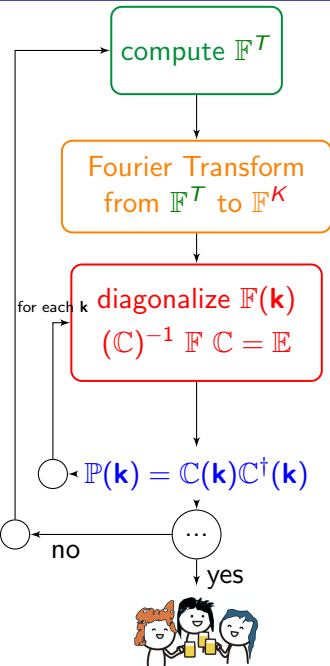
Fourier transform

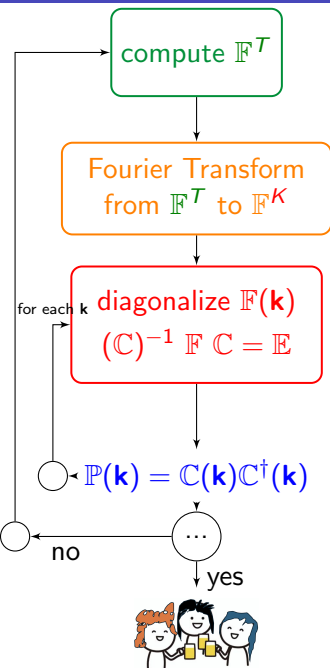
Exploit continuity

$$\mathbb{E} \rightarrow \mathbb{E}(\mathbf{k})$$

$$\mathbb{F} \rightarrow \mathbb{F}(\mathbf{k}) \Rightarrow \mathbb{C}(\mathbf{k})$$

by sampling Fock matrix in the reciprocal space



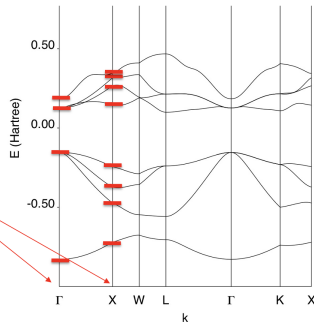
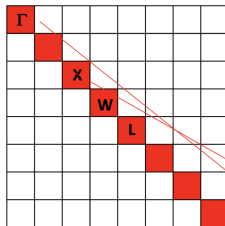


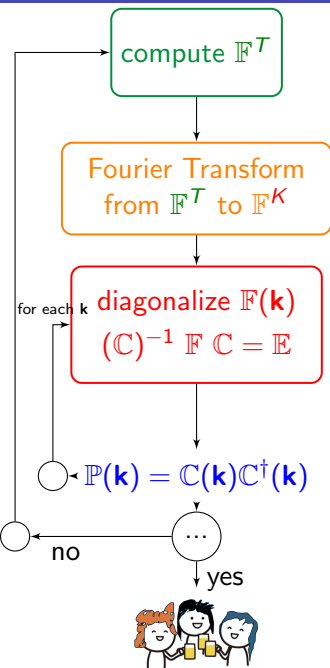
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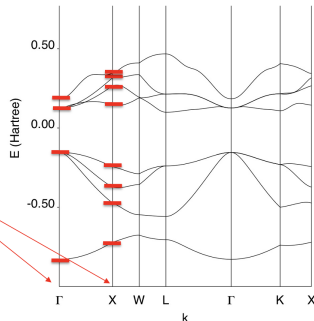
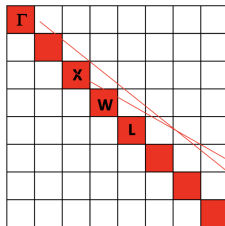


Exploit continuity

$$\mathbb{E} \rightarrow \mathbb{E}(\mathbf{k})$$

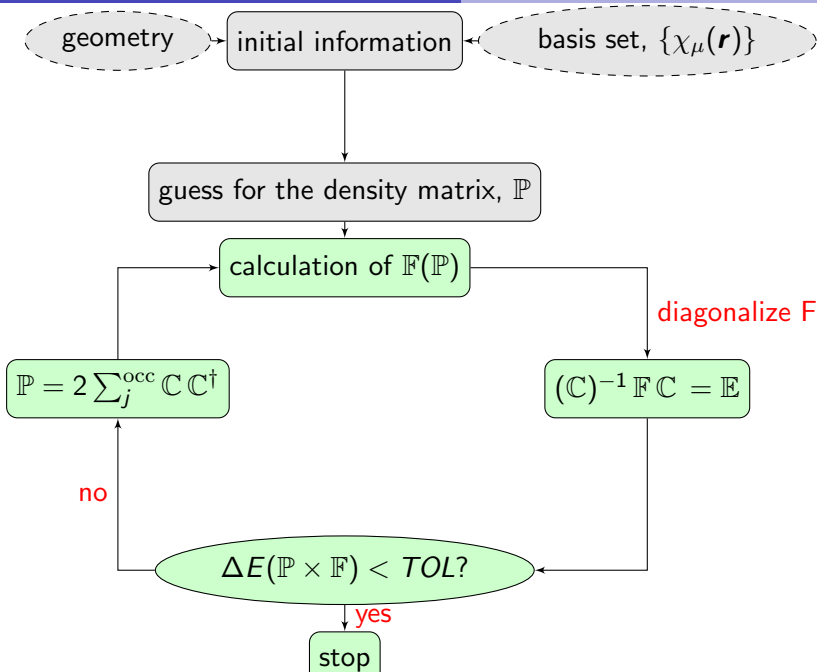
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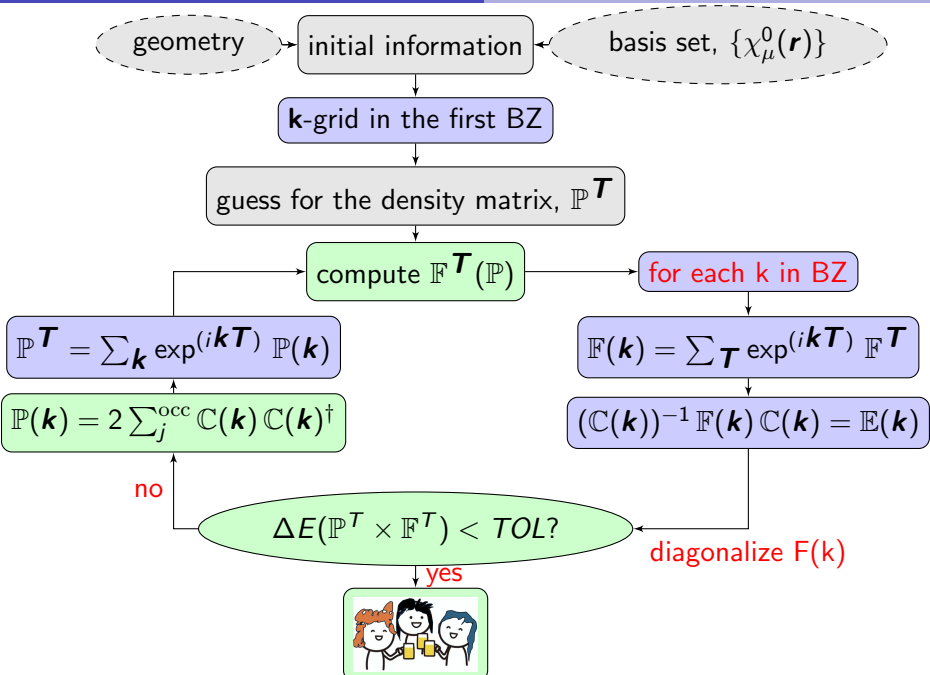
by sampling Fock matrix in the reciprocal space

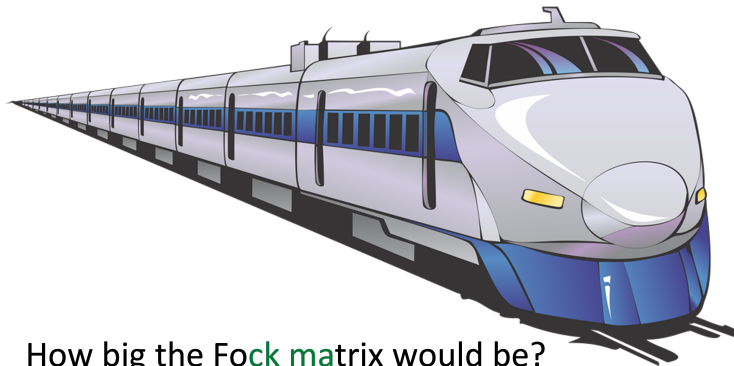


and anti-Fourier transform $\mathbb{P}$ to direct space

$$\mathbb{F}^T \Leftarrow \mathbb{P}^T = \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{r}} \mathbb{P}^{\mathbf{k}}$$







How big the Fock matrix would be?

IT DOESN'T MATTER! 😊😊😊

ACKNOWLEDGMENTS

I would like to thank: the whole CRYSTAL group



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and invite all of you at Platform 9 and $3/4$ where some strange boundary conditions are waiting for us...😊😊

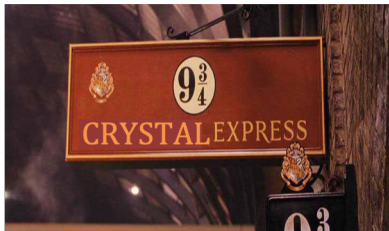


ACKNOWLEDGMENTS

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Sumud means: to resist, to face any type of adversity with obstinacy.

We are not the world's powerful,
but we are not powerless either.
We can use our voices to spread
their stories.

We can use our eyes and not look
away from this genocide,
and we can use our bodies, our
names, our privileges to do
something concrete,
because every sign, every word,
every gesture can make a
difference.

Even a single minute of silence.

